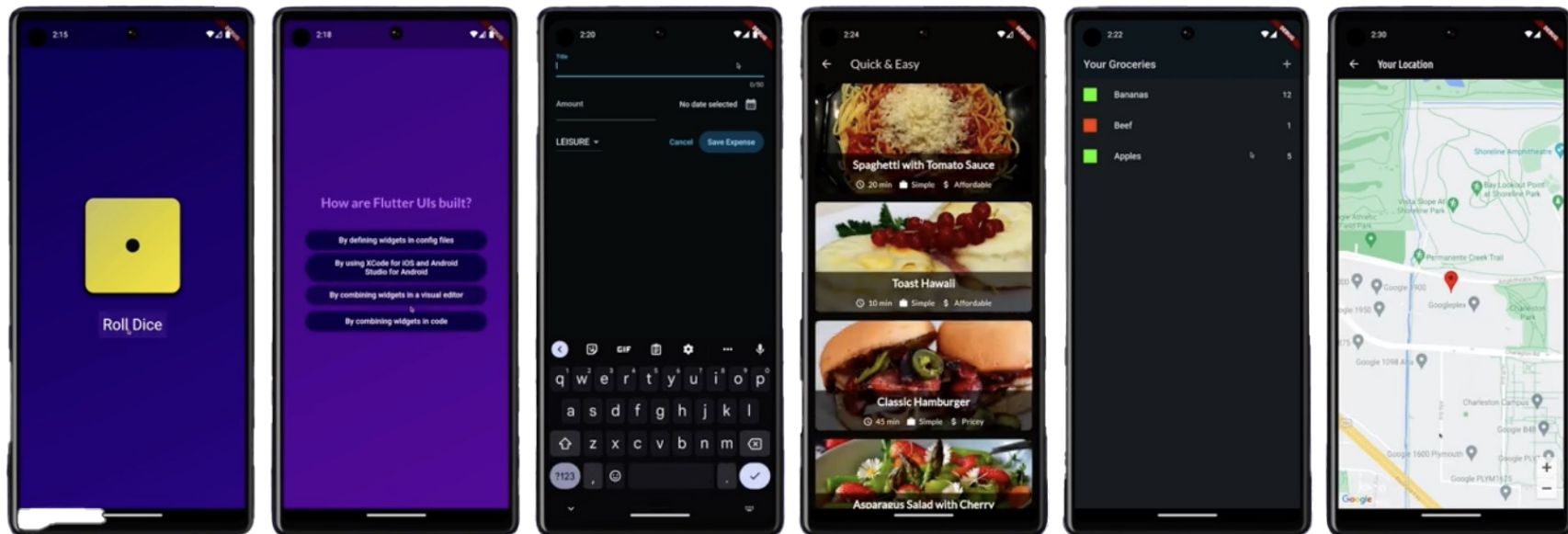


Software Design & Studio

Shan-Hung Wu
CS, NTHU

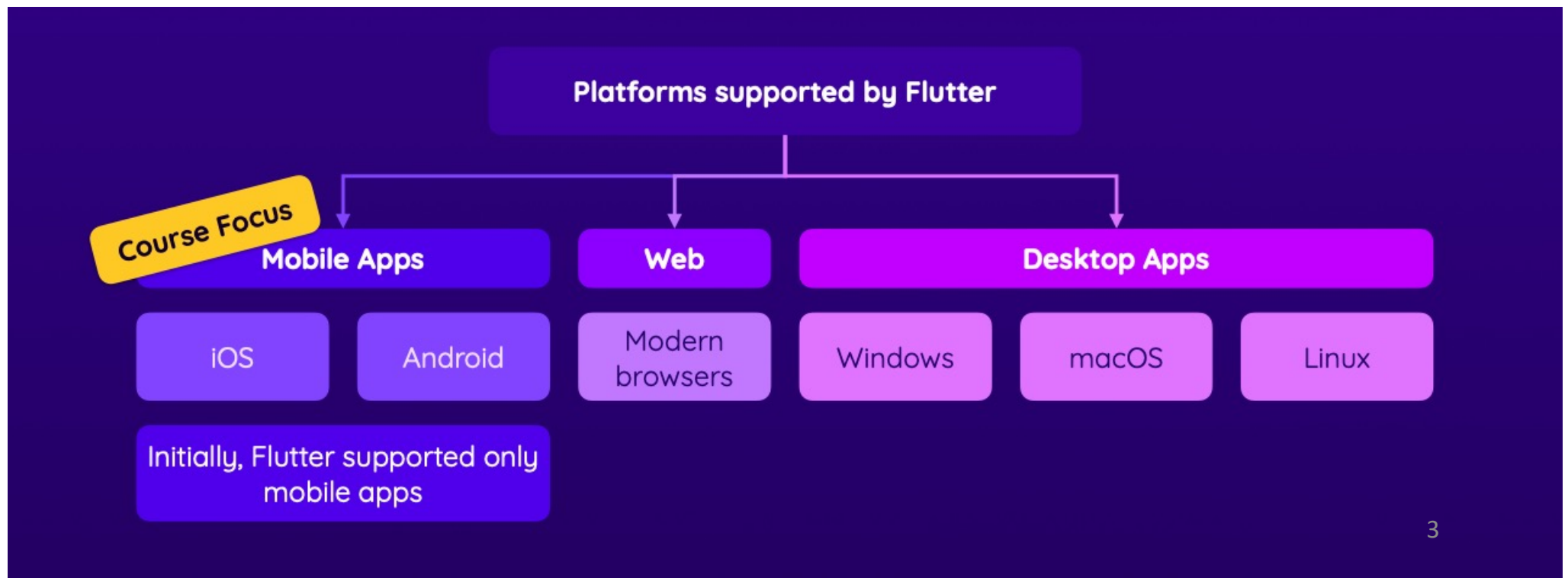
Course Objective

- To offer hand-on guidance in
 - Software *design* process
 - *Project-* and *team-based* software development
- Through writing intelligent apps:



Flutter for App Development

- Supports multiple platforms & screens
 - Shared codebase written in *Dart*
- Hot reload & high performance (60+ fps)



Why Dart?

- Easy to learn for people familiar with C, C# or Java
- New language features
 - Functional programming
 - Asynchronous programming
 - Null-safety
- Similar to JavaScript, but no legacy “bad” part

Flutter vs. Dart

- Dart: a programming language
- Flutter: libraries, framework and tools
- You write code in Dart within Flutter's framework
- Then, Flutter tools compile it to native code for each of your target platforms

Software Design

- Term project: a *useful* app empowered by *AI* technologies



- Usefulness through design sprint
 - Problem identification, solution ideation, competitive analysis, prototyping and usability testing
- AI technologies
 - Machine Learning & Deep Learning concepts, generative AI and APIs

Interleaved Sections

1. Flutter and Dart
 2. AI fundamentals and services
 3. Software design process and user-centric development
- [Syllabus](#) (subject to change)

Classes

- Tue: lectures
 - at Delta 105
- Thu: labs
 - at EECS 326 & 328
- Course website: www.cs.nthu.edu.tw/~shwu

Grading (1/2)

- Labs: **30%**
- Design demo (with usability testing): **40%**
- Implementation demo: **30%**
- Q/A bonus: up to **5%**

Grading (2/2)

- **Late** lab submissions only get **60%** of original scores
- Demos get **60%** of original scores if ***all your team members rate you as “non-contributing”***

Questions?

FAQ (1/3)

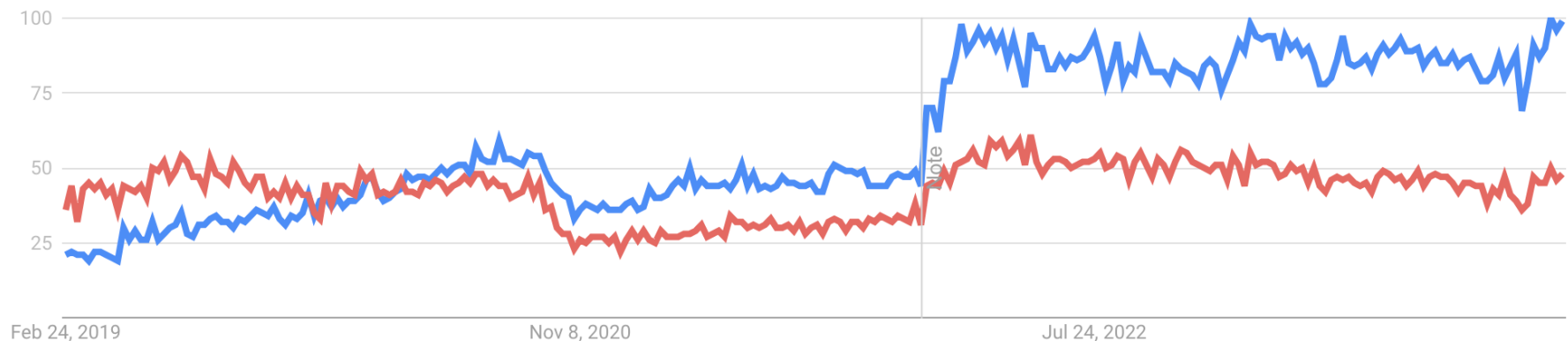
- Is this a light-loading class or heavy-loading class?
 - Should be heavy to most students
- Is this a programming language course?
 - No, we teach Dart in only 2 weeks
- Is this a software engineering (SE) course?
 - No. We don't focus on SE theories, but you will learn some "best practices"
- Is this an entrepreneur course?
 - No. We don't talk about how to make money

FAQ (2/3)

- How many people a team?
 - 4 to 5 people
 - Your apps must have a **mobile** frontend & **backend**
- What backend will we use
 - [Firebase](#), a Backend-as-a-Service (BaaS)
- Do we need to come to the class?
 - No, as long as you can pass
- Can I use generative AI to write code?
 - Fine, but make sure it helps you **learn** rather than score

FAQ (3/3)

- Are we going to interact with open-source software?
 - Yes, Flutter & Dart are open-source projects themselves
 - 3rd Dart/Flutter libraries through pub.dev
- Why not JavaScript + React + React Native?



TODO

- Complete these tutorials by Thu:
 - [Environment setup](#)
 - [Write your first Flutter app](#)